

PARC NATL DES PINGUALUIT, QC, Canada

WMO: 719760

Lat: 61.313N

Lon: 73.668W

Elev: 503

StdP: 95.43

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.5	-38.3	-44.5	0.0	-40.0	-42.1	0.1	-38.0	15.1	-17.9	12.6	-21.5	1.5	250	1.061

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	20.5	13.7	17.5	11.6	15.3	10.9	14.6	18.9	12.9	16.0	11.4	14.5	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	9.6	15.9	10.9	8.6	14.1	9.7	7.9	13.0	42.5	18.8	37.8	16.1	33.9	14.6	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.6	12.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.6	-26.9	-27.8	-23.6	-15.5	-5.8	3.5	9.3	8.6	2.5	-5.8	-14.1	-20.8
	DBStd	14.18	7.26	7.00	6.65	5.79	4.79	5.49	3.97	3.63	3.61	4.62	6.23	6.17
	HDD10.0	7233	1146	1059	1041	764	489	209	61	69	227	489	723	956
	HDD18.3	10196	1404	1292	1299	1014	747	446	281	301	476	748	973	1214
	CDD10.0	82	0	0	0	0	0	15	40	27	1	0	0	0
	CDD18.3	3	0	0	0	0	0	2	2	0	0	0	0	0
	CDH23.3	35	0	0	0	0	0	16	19	0	0	0	0	0
	CDH26.7	8	0	0	0	0	0	4	4	0	0	0	0	0
Wind	WSAvg	5.5	4.8	3.3	5.3	5.9	6.2	5.6	5.6	6.3	6.6	6.2	5.3	5.3
Precipitation	PrecAvg	452	18	18	27	22	27	43	50	69	65	49	38	27
	PrecMax	604	56	51	105	55	52	84	86	126	116	79	68	49
	PrecMin	304	5	2	5	5	7	12	12	34	34	26	13	9
	PrecStd	73	11	12	20	12	11	18	16	23	18	14	15	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.7	-6.4	-4.4	0.1	8.3	25.7	26.5	21.4	16.1	4.7	0.3	-5.0
		MCWB	-1.3	-6.7	-4.6	0.0	4.8	16.6	16.5	14.3	10.5	4.3	0.1	-5.3
	2%	DB	-8.8	-10.7	-8.1	-2.0	4.8	19.1	21.5	18.3	11.6	2.2	-2.7	-7.5
		MCWB	-8.9	-11.3	-8.4	-2.1	2.7	12.1	14.6	12.5	8.6	1.9	-2.8	-7.8
	5%	DB	-13.0	-14.8	-10.9	-4.8	2.8	15.2	18.1	15.9	9.4	0.9	-4.3	-9.5
		MCWB	-13.1	-15.1	-11.2	-5.2	1.4	9.4	12.2	11.7	7.7	0.6	-4.5	-9.8
	10%	DB	-15.7	-17.4	-13.7	-6.6	1.0	12.0	15.8	14.1	7.5	-0.2	-6.4	-12.1
		MCWB	-15.9	-17.7	-13.9	-7.0	0.3	8.0	11.1	10.8	6.0	-0.4	-6.6	-12.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.0	-8.0	-5.8	-0.4	4.0	16.3	18.0	15.6	10.3	4.8	-0.8	-5.1
		MCDB	-7.0	-7.9	-5.5	-0.4	5.8	25.1	25.7	18.6	11.9	5.1	-0.8	-4.9
	2%	WB	-11.8	-12.2	-8.6	-2.7	2.0	12.4	15.2	13.6	9.0	2.0	-2.7	-7.7
		MCDB	-11.6	-11.7	-8.4	-2.6	3.2	18.5	20.7	17.0	10.0	2.2	-2.6	-7.5
	5%	WB	-14.2	-15.6	-11.2	-5.5	0.8	9.8	13.4	12.3	7.0	0.5	-4.2	-9.8
		MCDB	-14.1	-15.4	-10.9	-5.2	1.6	14.6	16.6	15.0	8.5	0.6	-4.1	-9.5
	10%	WB	-16.7	-18.1	-13.6	-7.5	-0.4	7.9	11.9	11.1	5.7	-0.6	-6.2	-12.4
		MCDB	-16.4	-17.9	-13.3	-7.1	0.2	11.9	15.4	13.7	6.8	-0.4	-6.0	-12.1
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.1	7.8	8.1	6.1	6.5	7.3	5.9	4.9	4.1	5.8	6.0
		MCDBR	8.6	8.0	9.1	9.7	8.6	12.0	11.1	8.7	8.9	4.2	5.8	7.1
		MCWBR	7.0	7.5	8.3	8.3	6.0	5.9	5.1	4.5	5.5	4.4	5.8	7.1
	5% WB	MCDBR	6.7	7.2	8.5	8.3	6.8	11.4	9.8	7.4	7.3	4.2	6.0	7.1
		MCWBR	6.7	7.2	8.4	8.3	5.9	6.1	5.3	4.7	5.5	4.1	6.0	7.1
Clear-Sky Solar Irradiance	taub	0.191	0.233	0.273	0.341	0.341	0.330	0.333	0.325	0.320	0.269	0.202	0.158	
	taud	2.034	2.182	2.144	1.959	2.101	2.384	2.470	2.502	2.408	2.260	2.038	1.909	
	Ebn at Noon	566	771	847	835	873	897	886	866	796	703	523	425	
	Edh at Noon	38	65	98	146	138	107	96	85	78	60	36	24	
All-Sky Solar Radiation	RadAvg	0.37	1.30	3.20	5.52	6.66	5.52	4.88	3.67	2.11	0.81	0.32	0.17	
	RadStd	0.03	0.07	0.10	0.19	0.50	0.62	0.61	0.36	0.24	0.12	0.07	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page